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Professional Summary

- Results-driven Data Analyst/Engineer Professional with 5+ years of progressive experience designing, developing, and optimizing large-scale data systems across media, e-commerce, energy, and sports analytics domains.
- Demonstrated expertise in architecting end-to-end data pipelines for both batch and real-time applications, leveraging Python, SQL, DBT, Airflow, Spark, Kafka, AWS Glue, Athena, EMR, Azure Data Factory, and Databricks, ensuring high availability, scalability, and reliability of data infrastructure.
- Strong background in cloud-based architectures, with hands-on experience across AWS (Redshift, S3, Lambda, Kinesis, Glue, SageMaker, EMR), Azure (Data Lake Gen2, ADF, Spark), and GCP, enabling efficient ingestion, transformation, and orchestration of petabyte-scale datasets.
- Proven ability to partner with data science teams to productionize ML pipelines, integrating predictive models, recommendation engines, and classification workflows into robust, monitored production systems that support real-time decision-making.
- Adept at data modeling and optimization for large-scale analytical workloads and near-real-time reporting applications, with experience implementing streaming solutions (Kafka, Spark Streaming, HDFS, Pub/Sub, HBase) for customer segmentation, anomaly detection, and event-driven analytics.
- Delivered measurable impact, including:
 1. 40% faster data availability by building real-time ETL pipelines for player analytics in sports performance tracking.
 2. 60% reduction in manual reporting effort by automating dashboards and KPI generation across multiple business domains.
 3. 10% energy savings through SQL-based optimization of transformer load distribution in the energy sector.
- Leadership and collaboration: Worked closely with BI teams, backend engineers, and product managers to translate business goals into scalable data architectures. Mentored junior engineers on pipeline development, orchestration, and performance tuning, while serving as a thought partner to senior leadership on technology adoption.
- Strong communication and stakeholder engagement skills, with a track record of translating complex technical concepts into actionable insights for executives, coaches, scientists, and cross-functional teams.
- Passionate about innovation in data engineering—continuously exploring emerging tools such as Flink, Daft, Bedrock, and graph-based data processing to improve reliability, efficiency, and performance across large-scale pipelines.
- Educational foundation includes a Master's in Data Analytics Engineering (George Mason University) and Bachelor of Technology in Electrical Engineering, blending technical depth with analytical rigor to deliver robust, future-ready data systems.

Technical Skills

Programming & Scripting: Python (Pandas, NumPy, PySpark, Flask, FastAPI), SQL (T-SQL, PL/SQL, Snowflake SQL, HiveQL), R, MATLAB, C/C++, Java

Data Engineering & Processing: ETL/ELT development, Data Wrangling, Data Cleansing, SQL Optimization, PySpark, Hadoop, Databricks, Hive, HDFS, HBase, Real-time & Batch Processing

Cloud Platforms:

- AWS: Redshift, S3, Glue, Athena, Lambda, EMR, Kinesis, DynamoDB, SageMaker

- Azure: Data Factory, Data Lake Gen2, Synapse, Databricks, HDInsight
- GCP: BigQuery, Pub/Sub, Dataflow, Dataproc, Cloud Storage, Vertex AI

Workflow Orchestration & Automation: Apache Airflow, Control-M, Azure Data Factory, AWS Step Functions, Talend, Matillion, Luigi, Prefect

Streaming & Real-Time Frameworks: Apache Kafka, Spark Streaming, Flink, Pub/Sub, Kinesis

Machine Learning & AI Integration: Productionizing ML pipelines, Model Deployment & Monitoring (SageMaker, Vertex AI, Databricks ML), Feature Engineering Pipelines, Real-time Inference APIs

Data Modeling & Databases: MySQL, PostgreSQL, Oracle SQL, Snowflake, BigQuery, Redshift, DynamoDB, MongoDB

Visualization & BI Tools: Power BI, Tableau, QlikView, Looker, Excel (Advanced – VBA, Pivot Tables, Macros), KPI Dashboards, Real-time Monitoring Dashboards

DevOps & CI/CD: GitHub/GitLab, CloudFormation, CI/CD pipelines for Data & ML Workflows

Statistical & Analytical Techniques: Regression, Hypothesis Testing, Time Series Forecasting, Predictive Modeling, Anomaly Detection, A/B Testing

Education

Master of Science in Data Analytics Engineering, George Mason University, Fairfax, VA

Bachelor of Technology from Sreenidhi Institute of Science & Technology, Hyderabad, India

Professional Experience

Data Analyst | WCA, Sterling, VA | Jan 2025 – Present

Responsibilities:

- Designed and managed ETL pipelines with T-SQL, MySQL, and AWS Glue, reducing ingestion latency by 40% and enabling near real-time analytics for player performance.
- Built Tableau dashboards for 150+ athletes, monitoring training metrics, match stats, and injury patterns, driving data-backed coaching strategies.
- Automated weekly performance reports using Excel Macros and Tableau extracts, cutting manual reporting effort by 60%.
- Developed Python scripts to clean and integrate wearable and sensor-based datasets, achieving 98% accuracy in unified datasets.
- Integrated AWS S3, Lambda, and Power BI for scalable reporting on training attendance and load monitoring, enabling long-term player trend analysis.
- Collaborated with sports scientists to define new KPIs for injury prevention, player workload, and performance monitoring.
- Partnered with cross-functional teams (coaches, analysts, executives) to deliver personalized performance reports aligned with strategic goals.
- Leveraged Looker dashboards for high-level reporting, providing executives with strategic insights into academy-wide operations.
- Standardized data governance and validation checks, ensuring quality and consistency across training and performance datasets.
- Mentored junior analysts and interns on Python, SQL, and visualization best practices, instilling a data-first culture across teams.

Environment & Tools: Python, MySQL, T-SQL, Tableau, Power BI, Excel Macros, AWS S3, Glue, Lambda, Looker, JIRA

Data Engineer | Blackline | Feb 2023 – Dec 2024

Responsibilities:

- Architected enterprise-scale ETL/ELT pipelines using Azure Data Factory, Spark SQL, U-SQL, and Databricks, enabling ingestion and transformation of multi-terabyte datasets.
- Built real-time data ingestion frameworks with Apache Kafka, Hive, HDFS, and HBase, supporting both batch and streaming workloads.
- Designed and optimized PySpark transformation jobs and Databricks notebooks, improving pipeline scalability and cutting processing time by 35%.
- Migrated large workloads into Snowflake, optimizing schema design, clustering, and partitioning to boost performance and reduce storage costs.
- Implemented workflow orchestration using Airflow and Control-M, ensuring automated, reliable, and scalable pipeline execution.
- Partnered with BI and Product teams to develop executive dashboards in Tableau, Power BI, and QlikView, providing C-suite with actionable insights.
- Integrated CI/CD pipelines with GitLab and Jenkins for automated deployments and continuous delivery of production pipelines.
- Applied data governance frameworks and validation checks, improving reliability and preventing downstream errors.
- Enabled cross-team collaboration with product managers, backend engineers, and BI analysts, aligning data pipelines with business requirements.
- Supported cloud adoption initiatives, optimizing data lake operations in Azure Data Lake Gen2 while ensuring cost efficiency.
- Mentored junior engineers on pipeline design, orchestration, and coding best practices, reinforcing engineering excellence.

Environment & Tools: Apache Kafka, Azure Data Factory, Data Lake Gen2, Spark, Databricks, Hive, HDFS, HBase, SQL Server, Control-M, Python, Tableau, Power BI, QlikView, GitLab, Jenkins

Data Analyst | Viswam Power Technologies, Hyderabad, India | Jun 2021 – Dec 2022

Responsibilities:

- Automated data workflows using Python and VBA Macros, reducing processing time by 45% in energy reporting systems.
- Designed and implemented Power BI dashboards to monitor transformer load distribution and hotspot visualization, improving operational efficiency.
- Conducted SQL-based transformer performance analysis, cutting energy consumption costs by 10% through optimized load allocation.
- Built forecasting models with R for demand planning, improving resource allocation accuracy by 20%.
- Leveraged AWS Glue and Athena for scalable ETL and querying, reducing manual data handling and streamlining analytics.
- Transitioned legacy spreadsheet-based workflows into BI-driven dashboards, ensuring 97% reporting accuracy.
- Partnered with engineers to create data models that improved transformer maintenance scheduling and reduced downtime.
- Monitored data quality, governance, and validation rules, ensuring compliance with energy industry standards.
- Optimized database queries for PostgreSQL and Oracle SQL, improving system responsiveness.
- Collaborated with leadership to integrate insights into cost-saving initiatives, reducing operational expenditures.

Environment & Tools: Python, VBA Macros, SQL, R, Power BI, Excel, AWS Glue, Athena, PostgreSQL, Oracle SQL

Data Analyst | Flipkart | Aug 2019 – May 2021

Responsibilities:

- Developed ETL pipelines with Python, SQL, and AWS Glue, integrating datasets across MySQL, PostgreSQL, and DynamoDB for reporting.
- Designed Power BI and Tableau dashboards to track KPIs in sales, customer behavior, and business performance.
- Built complex SQL queries and audit reports, ensuring data quality and regulatory compliance across business units.
- Conducted predictive modeling and forecasting for revenue and customer retention using statistical techniques.
- Migrated reporting workloads to AWS Redshift, Athena, EMR, and S3, reducing query times by 50%.
- Built automated validation and data governance scripts in Python, reducing reporting errors and ensuring compliance.
- Supported business intelligence teams with tailored datasets for campaign analysis and performance reviews.
- Implemented Lambda-based automation workflows to streamline periodic data refresh cycles.
- Collaborated with cross-functional teams to deliver real-time reporting pipelines for logistics and operations.
- Improved reporting scalability by integrating event-driven architectures for faster sales data availability.

Environment & Tools: Python, SQL, AWS Glue, Redshift, Athena, EMR, S3, Lambda, Power BI, Tableau

Academic and Other Project Experience

SQL-Based Data Extraction & Reporting Optimization

- Designed advanced SQL queries (joins, CTEs, aggregations, window functions) to extract and analyze donor behavior trends across multiple campaigns.
- Built ETL pipelines with Python and cron jobs, automating data movement between MySQL and PostgreSQL, reducing processing time by 40%.
- Created segmentation models to classify donors by engagement and contribution frequency, improving campaign targeting and increasing conversion by 18%.
- Developed Tableau dashboards to provide executives with donor KPIs, contribution heatmaps, and campaign performance insights.
- Ensured financial data compliance through validation scripts and automated reconciliation checks across multiple data sources.
- Optimized query execution and indexing strategies, improving reporting performance by 50%.
- Partnered with fundraising teams to translate raw data into actionable donor outreach strategies.
- Established a reusable SQL code repository, improving reporting efficiency and knowledge transfer within the team.
- Designed a scalable data model to support cross-campaign analytics, enabling future growth of donor datasets.

Tools: SQL, Python, Tableau, MySQL, Excel

DNSSEC Outage Impact Analysis

- Queried 56+ billion DNS records using AWS Athena, EMR, and S3, enabling large-scale outage analysis across multiple regions.
- Implemented time-series anomaly detection in Python to identify DNSSEC failure points and outage propagation patterns.
- Developed region-based outage dashboards in Power BI, allowing stakeholders to visualize downtime patterns in real time.
- Designed ETL workflows using AWS Glue to pre-aggregate and structure log data for faster root-cause analysis.

- Built predictive analytics models to forecast high-risk regions based on outage history and anomaly scores.
- Optimized Athena query structures and partitions, reducing analysis costs by 30%.
- Partnered with network engineering teams to integrate insights into proactive outage mitigation strategies.
- Automated pipeline alerts via CloudWatch and Lambda, notifying teams of early signs of outages.
- Enhanced data storage efficiency by designing tiered data retention strategies in S3, reducing storage costs by 20%.

Tools: Python, AWS Athena, EMR, S3, Glue, Power BI

Facial Expression Recognition Using Deep Learning

- Collected and preprocessed large image datasets with OpenCV, handling augmentation, noise reduction, and image normalization.
- Built and trained CNN-based models using VGG-16, ResNet50, and MobileNet, optimizing hyperparameters for accuracy and efficiency.
- Developed a real-time GUI interface for webcam-based facial expression recognition, supporting instant classification.
- Achieved 92% accuracy with the optimized ResNet50 model, outperforming baseline implementations.
- Integrated the solution with TensorFlow and Scikit-learn pipelines for scalable deployment.
- Deployed the model as a Flask-based API, allowing integration with other applications.
- Implemented GPU-accelerated training on AWS EC2 instances, reducing training time by 50%.
- Designed automated model monitoring pipelines to track drift and maintain performance over time.
- Documented and versioned experiments using MLflow, ensuring reproducibility for future model improvements.

Tools: Python, TensorFlow, OpenCV, Scikit-learn

Real-Time ML Pipeline for Recommendation Engine

- Architected a real-time recommendation engine on GCP using Pub/Sub, Dataflow, BigQuery, and Vertex AI, enabling instant content personalization for a media client.
- Ingested 100M+ daily user activity events with Pub/Sub and processed streaming data using Apache Flink and Spark Structured Streaming.
- Designed ELT workflows in DBT for BigQuery to support analytics and feature engineering at scale.
- Partnered with data scientists to productionize ML models for user segmentation and content ranking, achieving 30% improvement in engagement rates.
- Implemented CI/CD pipelines with GitLab and Docker for model deployment and API integration.
- Created real-time monitoring dashboards in Looker and Grafana, tracking latency, model performance, and data quality SLAs.
- Built automated retraining workflows using Vertex AI Pipelines, ensuring model freshness and preventing drift.
- Optimized system reliability with low-latency architectural design, achieving sub-second inference response times.
- Collaborated with BI, backend engineering, and product teams, ensuring seamless integration of the recommendation engine into user-facing applications.

Tools: Google Pub/Sub, Apache Kafka, Apache Flink, Spark Structured Streaming, PySpark, BigQuery, Airflow, Looker